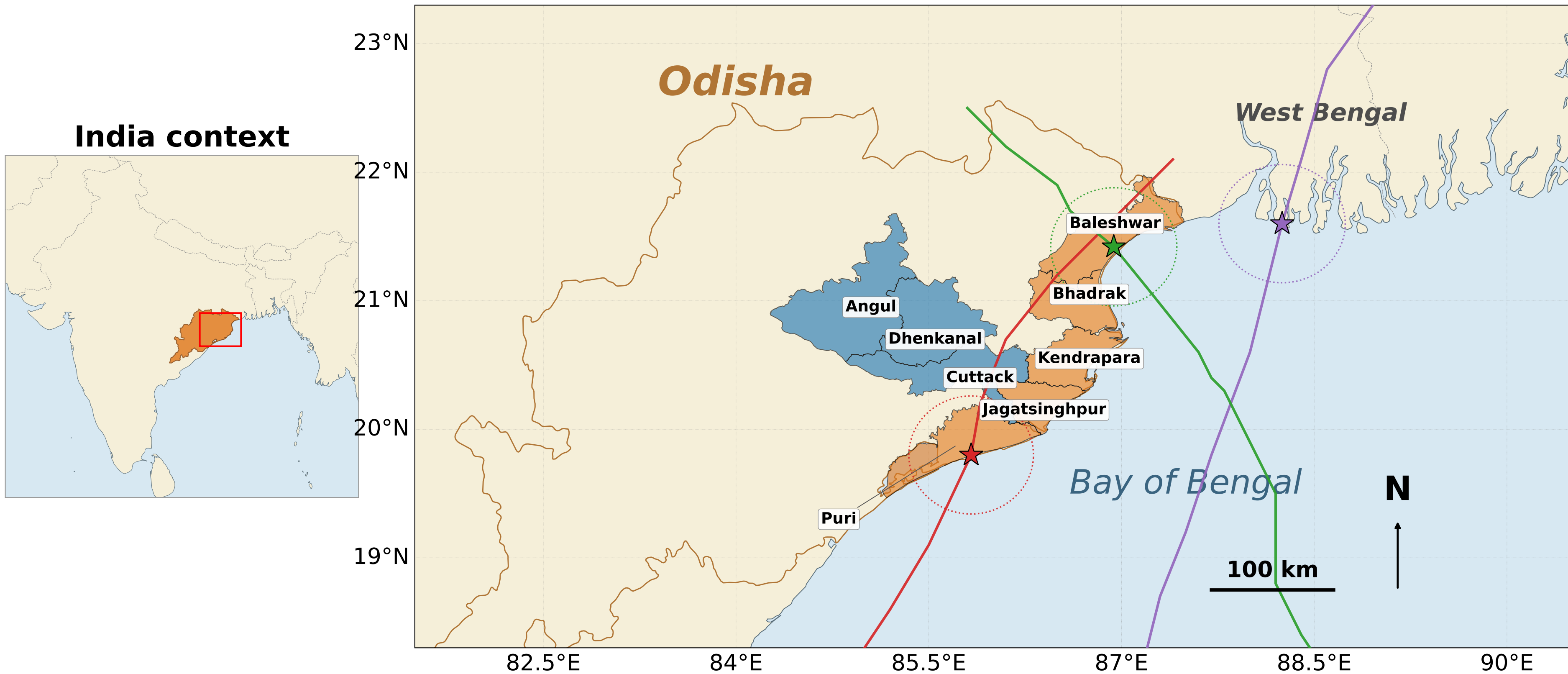


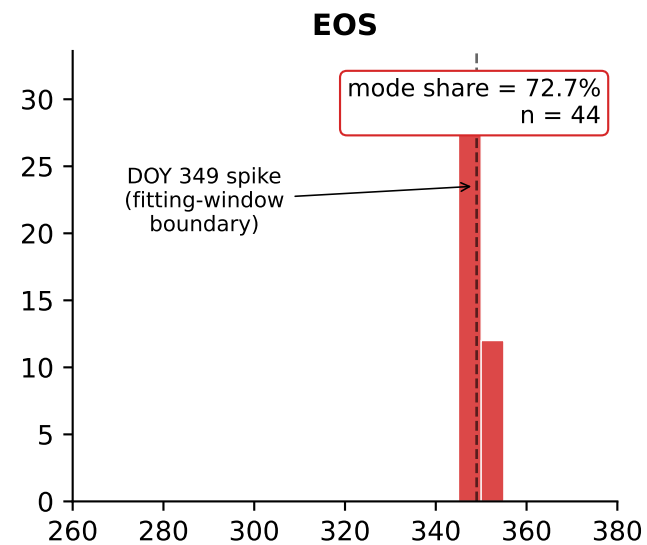
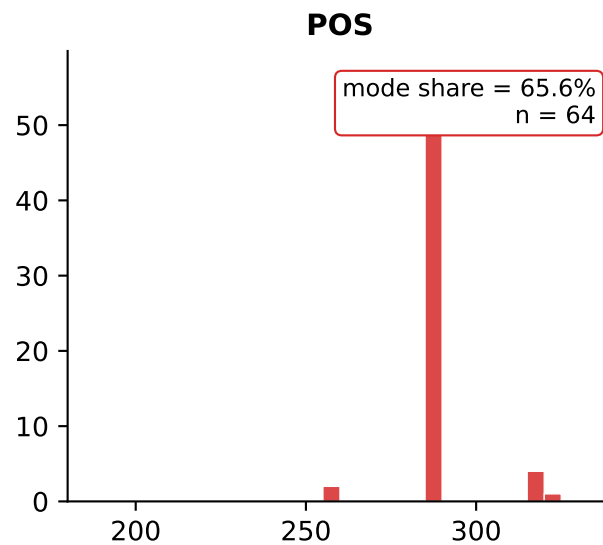
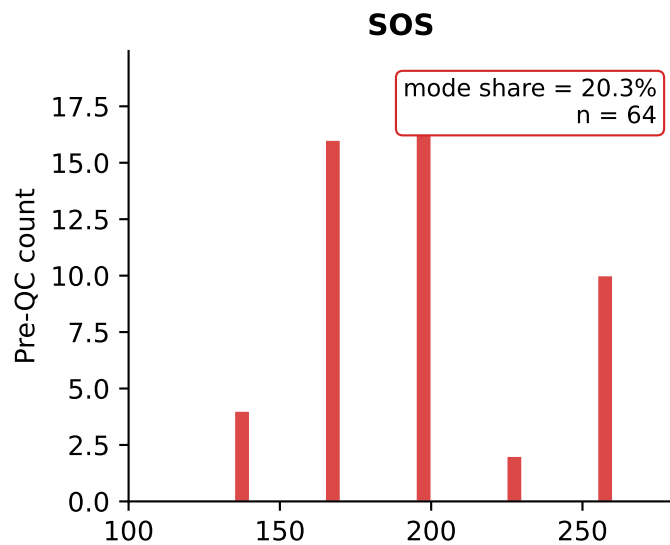


Legend

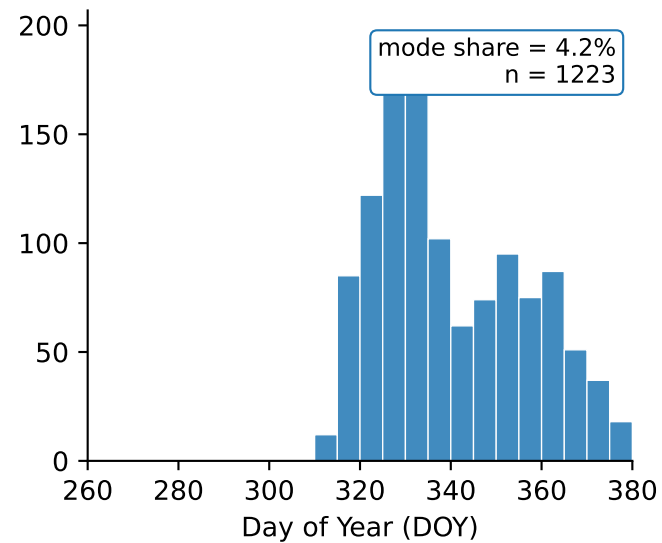
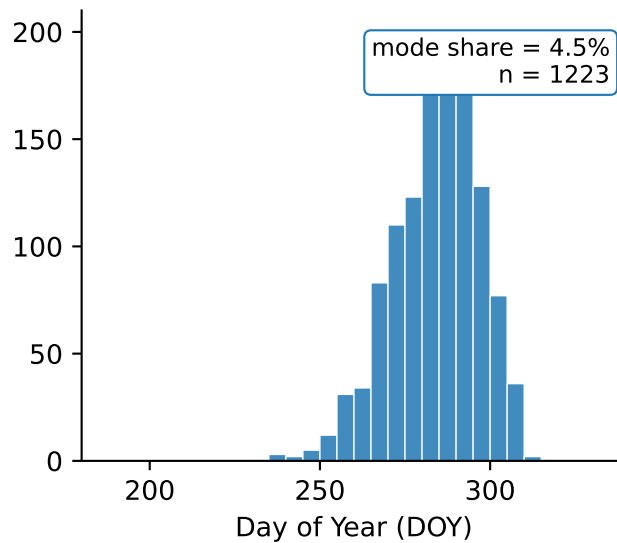
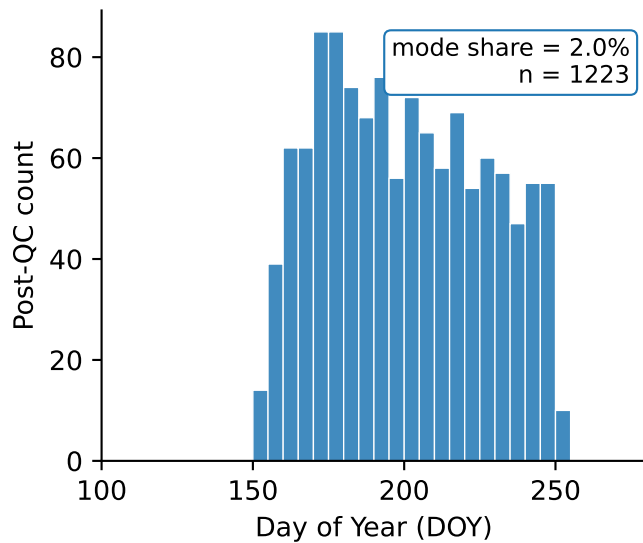
- Coastal treatment district (n=5)
- Inland control district (n=3)
- Cyclone Fani (3 May 2019, ESCS)
- Cyclone Amphan (20 May 2020, SuCS)
- Cyclone Yaas (26 May 2021, VSCS)
- ★ IMD-documented landfall site
- 50-km landfall buffer

Phenometric distributions before and after QC framework application

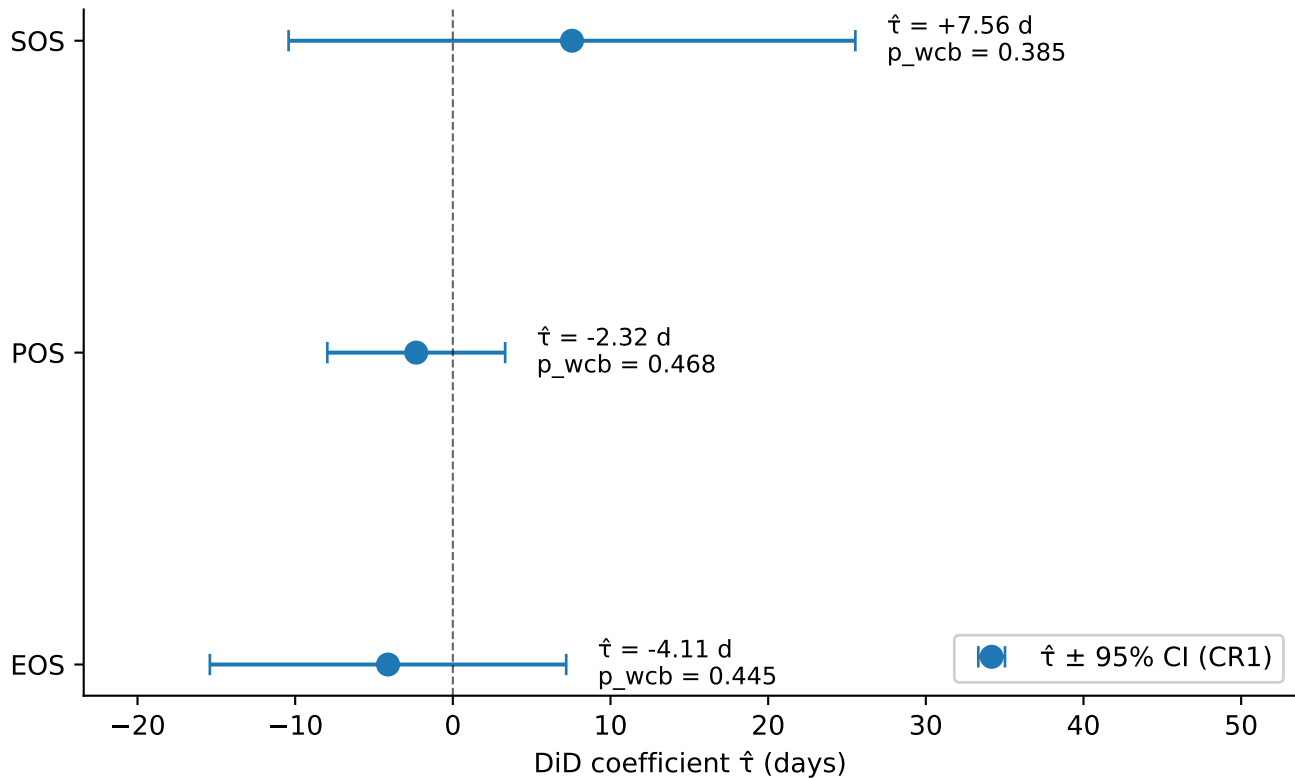
Pre-QC (v1.0.2)
district aggregates



Post-QC (v2.0)
cell-level fits

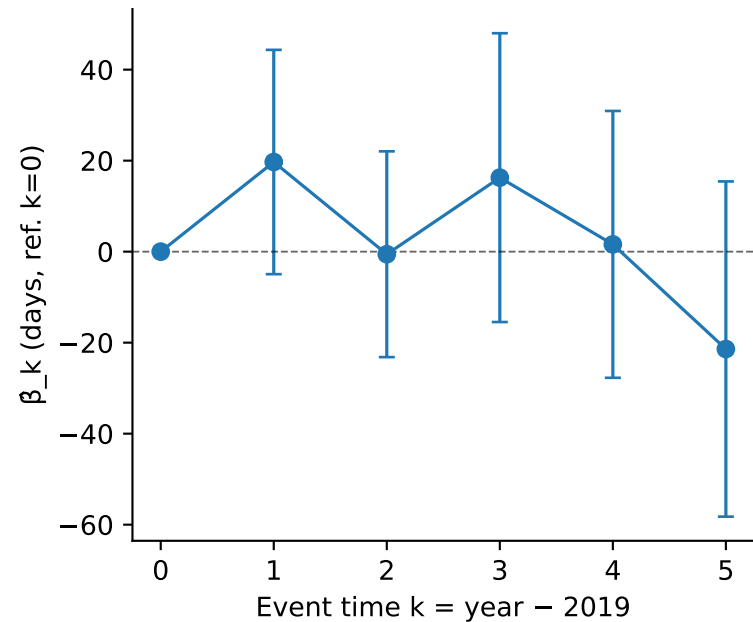


Static DiD on QC-cleaned v2.0 panel (n=48; 5 coastal + 3 inland)

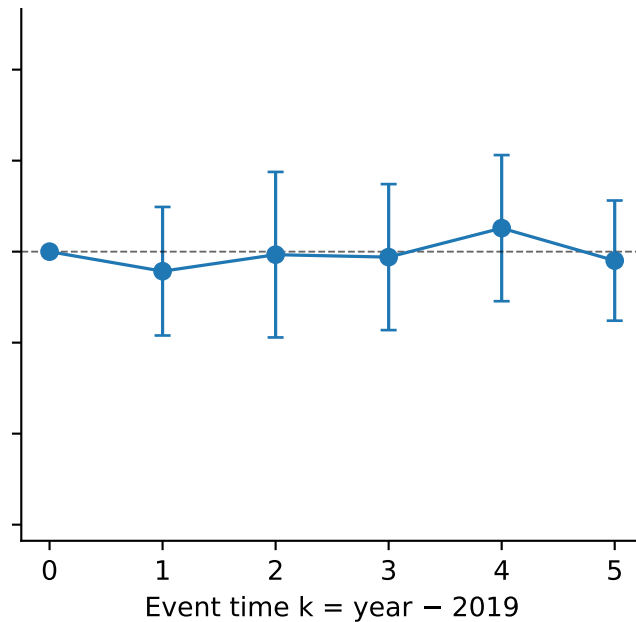


Event-study dynamics on QC-cleaned v2.0 panel (treatment districts only)

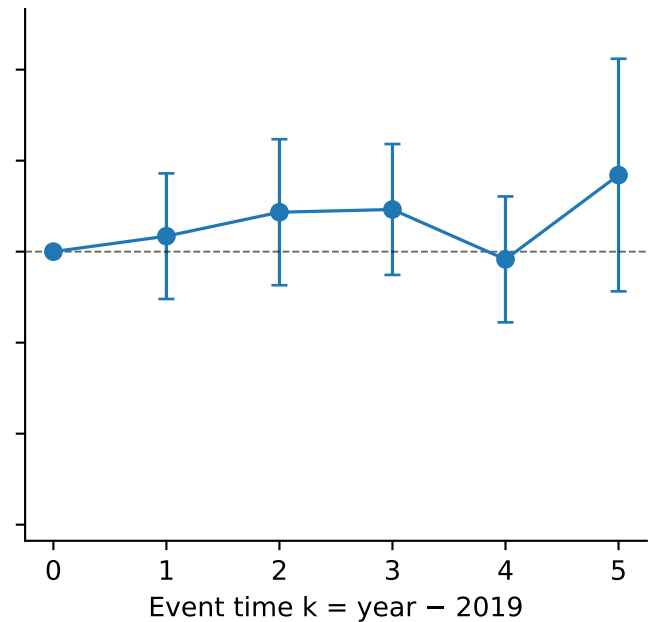
SOS



POS



EOS



Reproducible QC framework for Sentinel-2 phenology retrieval

Sentinel-2 L2A NDVI time series
(cell-level, Kharif window)

Stage A — Dekadal compositing
(10-day max-NDVI; cloud-screened L2A)

Stage B — Double-logistic fit
(Whittaker smooth → 6-parameter fit)

QC FRAMEWORK (this paper's contribution)

Gate A

Mode-share threshold

mode-share ≤ 0.20
(per phenometric,
panel-wide)

Fails in v1.0.2:
EOS 0.727 POS 0.656
SOS 0.203

Gate B

Biological plausibility

SOS $\in [\text{DOY } 155, 240]$
POS $\in [\text{DOY } 240, 320]$
EOS $\in [\text{DOY } 280, 380]$

(district-year
medians)

Gate C

Fit-quality screening

RMSR ≤ 0.15

fraction of pixels
with poor fit ≤ 0.30

per district-year cell

All three gates passed?

yes
PASS → Cell enters DiD
estimation panel (n = 48)

no
FAIL → Cell excluded
(20 EOS rows censored in v1.0.2)

Empirical validation on Odisha panel (2019–2024, 8 districts × 6 years)

EOS mode-share
72.7% → 8.3%

POS mode-share
65.6% → 8.3%

Unique EOS DOY values
2 → 38

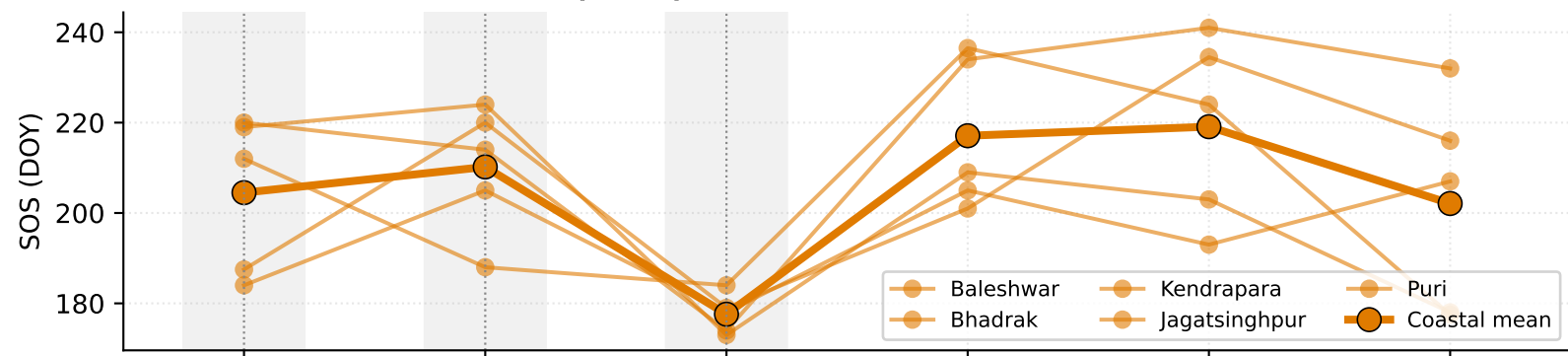
District-year median SOS, 2019-2024 (v2.0 QC-passing panel)

Fani

Amphan

Yaas

Coastal treatment districts (n = 5)



Inland control districts (n = 3)

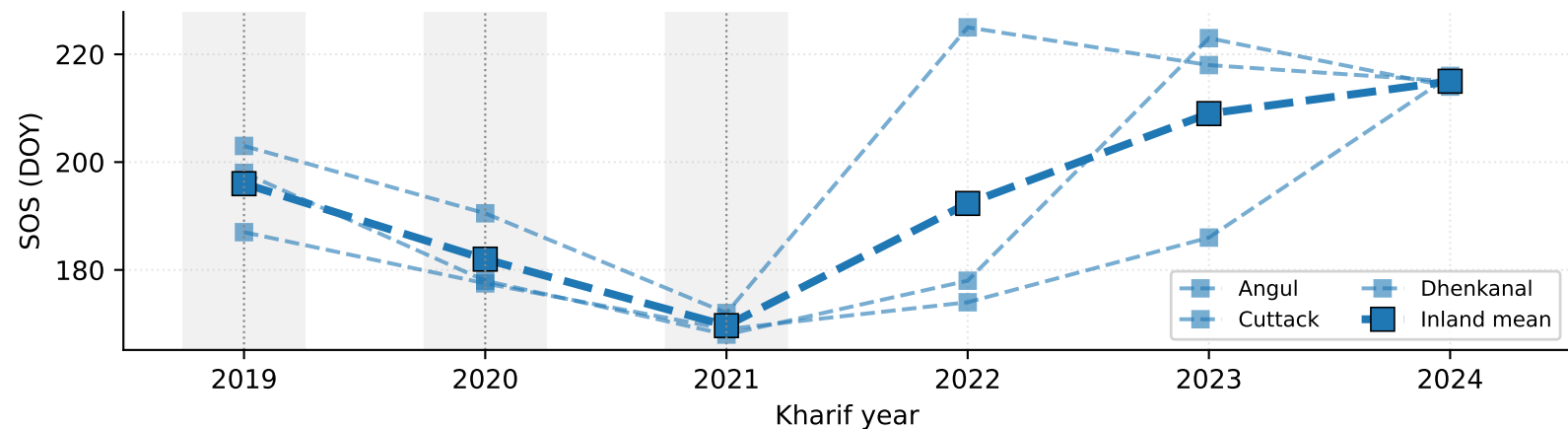
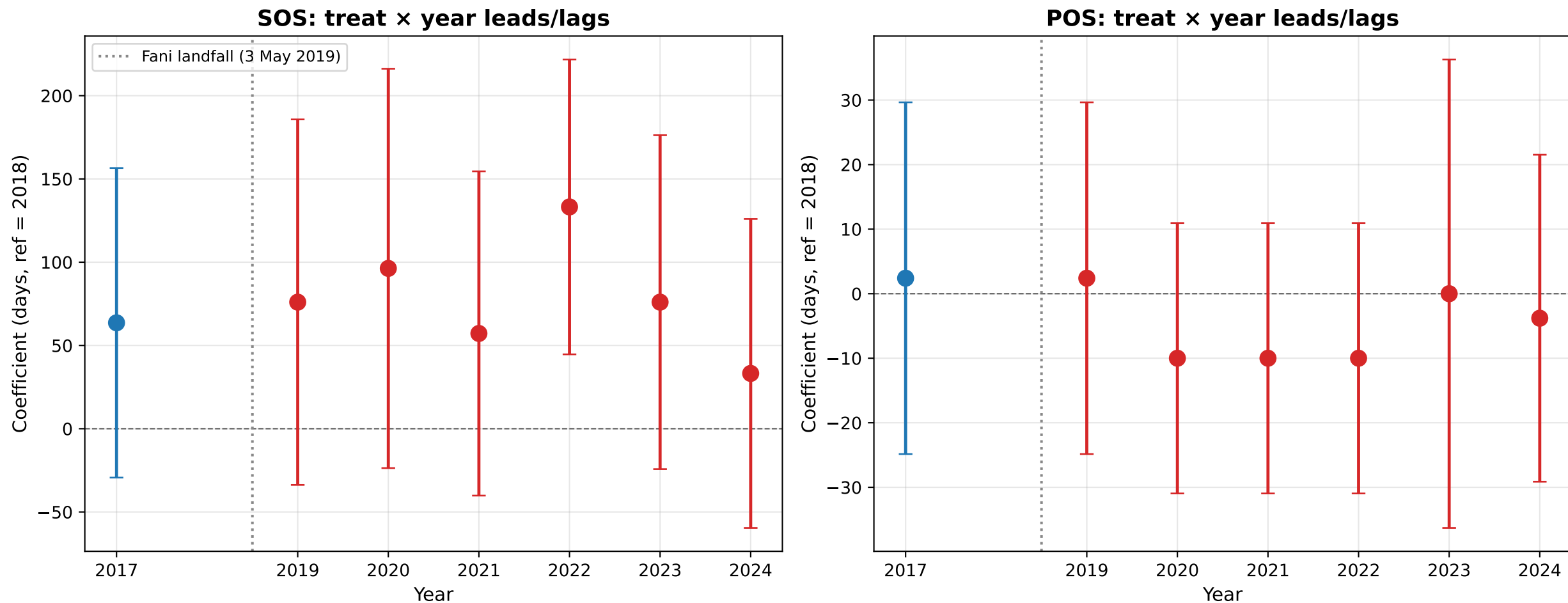
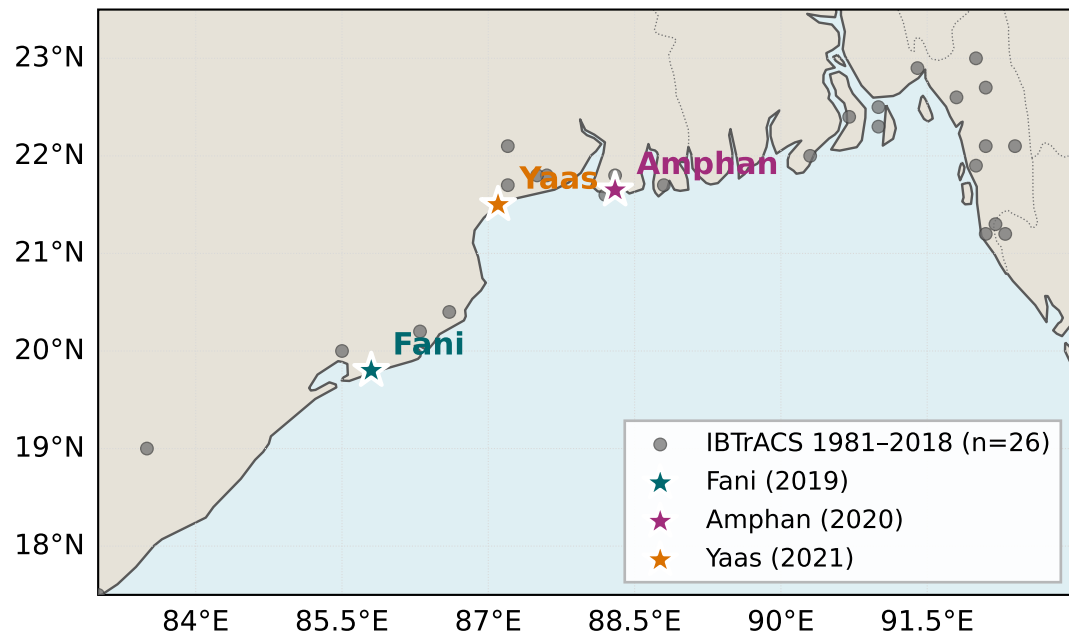


Figure 7. Event-study parallel-trends test (2017 pre-period lead, 2019-2024 post-period lags; ref = 2018)

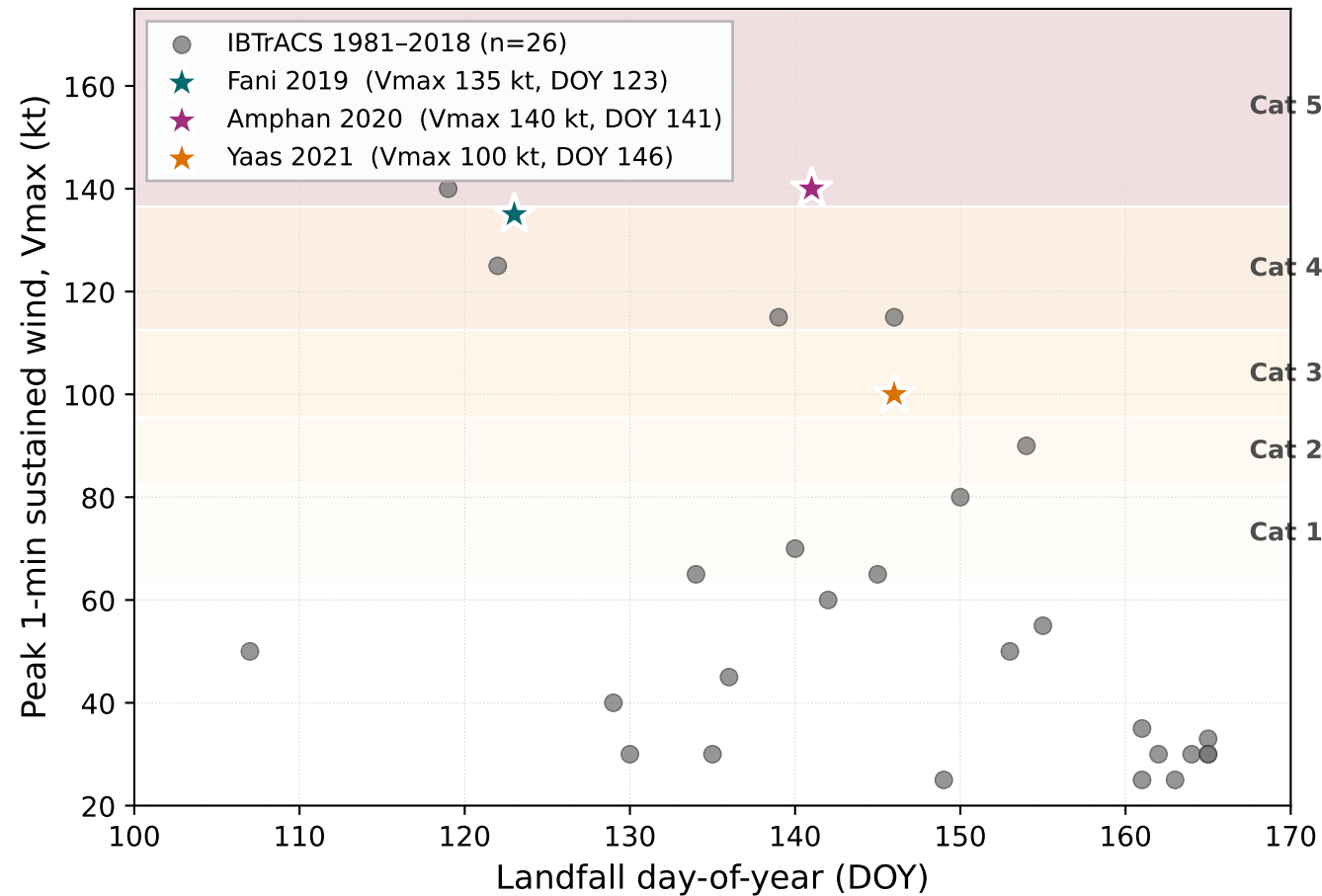


Note: EOS panel omitted — within-year variance is zero (349/350 DOY boundary artifact); no event-study coefficient identifiable. Landsat-8 2015–2018 extension (Module 13) provides additional pre-period leads.

(A) Bay of Bengal pre-Kharif landfalls

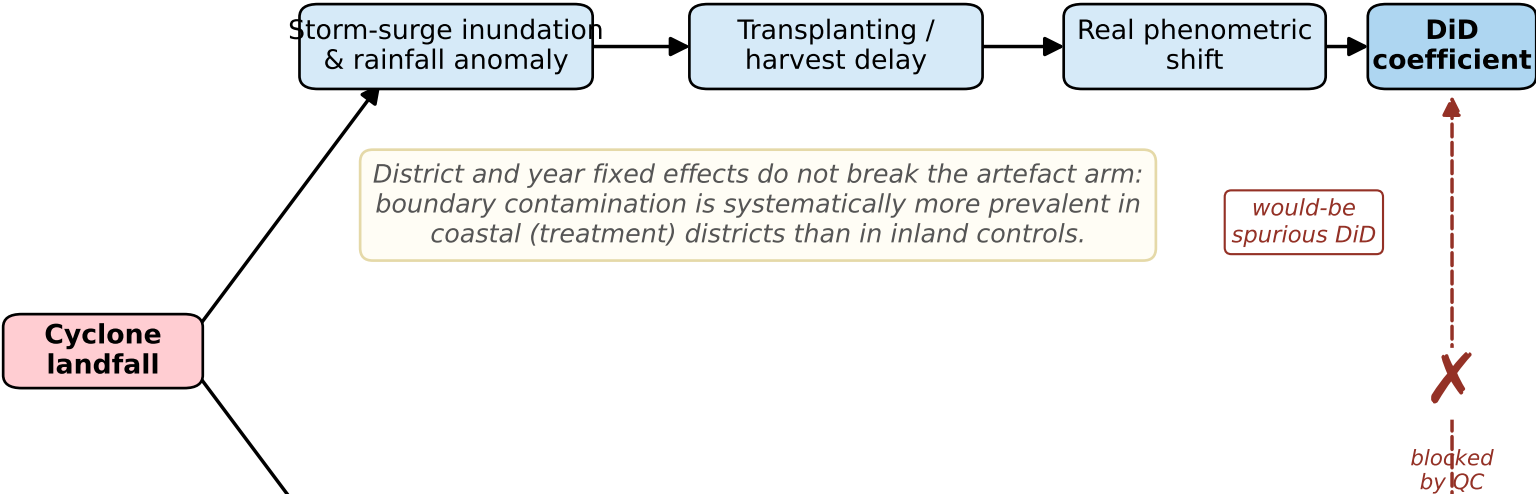


(B) Pre-Kharif landfall intensity vs. day-of-year



Identification DAG: boundary-quantisation artefact and the QC intercept

Pathway 1 — Genuine agronomic effect



Pathway 2 — Boundary-quantisation artefact (intercepted by QC)

